Exp. No. 7

Write a C program to find FIRST( ) - predictive parser for the given grammar

main.c

Run ▶

```c
#include<stdio.h>
#include<ctype.h>
void FIRST(char[],char );
void addToResultSet(char[],char);
int numOfProductions;
char productionSet[10][10];
int main()
{
int i;
char choice;
char c;
char result[20];
printf("How many number of productions ? :");
scanf(" %d",&numOfProductions);
for(i=0;i<numOfProductions;i++)
{
printf("Enter productions Number %d : ",i+1);
scanf(" %s",productionSet[i]);
}
do
{
printf("\n Find the FIRST of :");
scanf(" %c",&c);
FIRST(result,c);
printf("\n FIRST(%c)= { ",c);
for(i=0;result[i]!='\0';i++)
```

Output

```
How many number of productions ? :4
Enter productions Number 1 : S=AaAb
Enter productions Number 2 : S=BbBa
Enter productions Number 3 : A=$
Enter productions Number 4 : B=$

 Find the FIRST of :s

 FIRST(s)= {  s }
press 'y' to continue :
```

```c
printf(" %c ",result[i]);
printf("}\n");
printf("press 'y' to continue : ");
scanf(" %c",&choice);
}
while(choice=='y'||choice =='Y');
}
void FIRST(char* Result,char c)
{…}
void addToResultSet(char Result[],char val)
{
int k;
for(k=0 ;Result[k]!='\0';k++)
if(Result[k]==val)
return;
Result[k]=val;
Result[k+1]='\0';
}
```